

| DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE                                                                                                     |                                                                                                   |                                               |                                             |                                    |
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| Supplementary Winter Examination – 2024                                                                                                                     |                                                                                                   |                                               |                                             |                                    |
| Course: Third Year B. Tech. Branch: Electrical Engineering and Allied                                                                                       |                                                                                                   |                                               | Semester: VI                                |                                    |
| Subject Code & Name: BTEOE605B - Power Plant Engineering                                                                                                    |                                                                                                   |                                               |                                             |                                    |
| Max Marks: 60                                                                                                                                               |                                                                                                   | Date: 30/12/2024                              |                                             | Duration: 3 Hr.                    |
| <b>Instructions to the Students:</b>                                                                                                                        |                                                                                                   |                                               |                                             |                                    |
| 1. Each question carries 12 marks.                                                                                                                          |                                                                                                   |                                               |                                             |                                    |
| 2. Question No. 1 will be compulsory and include objective-type questions.                                                                                  |                                                                                                   |                                               |                                             |                                    |
| 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.                                                             |                                                                                                   |                                               |                                             |                                    |
| 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question. |                                                                                                   |                                               |                                             |                                    |
| 5. Use of non-programmable scientific calculators is allowed.                                                                                               |                                                                                                   |                                               |                                             |                                    |
| 6. Assume suitable data wherever necessary and mention it clearly.                                                                                          |                                                                                                   |                                               |                                             |                                    |
|                                                                                                                                                             |                                                                                                   |                                               | (Level/CO )                                 | Marks                              |
| <b>Q. 1</b>                                                                                                                                                 | <b>Objective type questions. (Compulsory Question)</b>                                            |                                               |                                             | <b>12</b>                          |
| 1                                                                                                                                                           | In a nuclear power plant, the energy is produced through the process of:                          |                                               |                                             | 1                                  |
|                                                                                                                                                             | a. Combustion                                                                                     | b. Fission                                    | c. Fusion                                   | d. Condensation                    |
| 2                                                                                                                                                           | The efficiency of a thermal power plant is primarily affected by:                                 |                                               |                                             | 1                                  |
|                                                                                                                                                             | a. The type of fuel used.                                                                         | b. The temperature of the steam.              | c. The design of the boiler.                | d. All of the above                |
| 3                                                                                                                                                           | The efficiency of a hydroelectric power plant mainly depends on:                                  |                                               |                                             | 1                                  |
|                                                                                                                                                             | a. Water flow rate.                                                                               | b. The height of the water source.            | c. The type of turbine used.                | d. All of the above                |
| 4                                                                                                                                                           | Which of the following is a common method used to improve the efficiency of thermal power plants? |                                               |                                             | 1                                  |
|                                                                                                                                                             | a. Using superheated steam                                                                        | b. Reducing the steam pressure                | c. Decreasing the temperature of the boiler | d. Increasing the fuel consumption |
|                                                                                                                                                             | The major environmental concerns associated with thermal power plants include:                    |                                               |                                             | 1                                  |
|                                                                                                                                                             | a. Noise pollution                                                                                | b. Greenhouse gas emissions and air pollution | c. Habitat destruction                      | d. Water scarcity                  |
| 6                                                                                                                                                           | Which of the following is a major limitation of hydroelectric power plants?                       |                                               |                                             | 1                                  |
|                                                                                                                                                             | a. High operating costs                                                                           | b. Limited by geographical location           | c. Environmental pollution                  | d. High fuel costs                 |

|     |                                                                                                                                        |                                                       |                                                        |                                                                        |  |    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------|--|----|
| 7   | What is the primary function of the dam in a hydroelectric power plant?                                                                |                                                       |                                                        |                                                                        |  | 1  |
|     | a. Store water and create a reservoir                                                                                                  | b. Generate electrical energy                         | c. Control water temperature                           | d. Convert electrical energy to mechanical energy                      |  |    |
| 8   | What is the primary function of the control rods in a nuclear reactor?                                                                 |                                                       |                                                        |                                                                        |  | 1  |
|     | a. Absorb neutrons and control the rate of fission                                                                                     | b. Increase the temperature of the reactor            | c. Cool the reactor                                    | d. Generate electricity                                                |  |    |
| 9   | The by-products of nuclear fission are:                                                                                                |                                                       |                                                        |                                                                        |  | 1  |
|     | a. Water and carbon dioxide                                                                                                            | b. Neutrons and energy                                | c. Radioactive isotopes and energy                     | d. Heat and light                                                      |  |    |
| 10  | The combined operation of power plants refers to:                                                                                      |                                                       |                                                        |                                                                        |  | 1  |
|     | a. The operation of different types of power plants together to meet the power demand                                                  | b. The use of a single fuel for multiple power plants | c. The process of maintaining the plants independently | d. The combination of fossil and renewable energy sources in one plant |  |    |
| 11  | Which of the following types of power plants are typically combined for optimal operation?                                             |                                                       |                                                        |                                                                        |  | 1  |
|     | a. Thermal and hydroelectric power plants                                                                                              | b. Nuclear and thermal power plants                   | c. Wind and solar power plants                         | d. Thermal, hydroelectric, and nuclear power plants                    |  |    |
| 12  | The main environmental benefit of using renewable energy sources is:                                                                   |                                                       |                                                        |                                                                        |  | 1  |
|     | a. They reduce greenhouse gas emissions                                                                                                | b. They are more cost-effective than fossil fuels     | c. They do not require any equipment                   | d. They are available all the time                                     |  |    |
|     |                                                                                                                                        |                                                       |                                                        |                                                                        |  |    |
| Q.2 | Solve the following.                                                                                                                   |                                                       |                                                        |                                                                        |  | 12 |
| A)  | Discuss the key factors involved in the selection of a site for thermal steam power plants.                                            |                                                       |                                                        |                                                                        |  | 6  |
| B)  | Explain with schematic diagram of a typical hydroelectric power plant                                                                  |                                                       |                                                        |                                                                        |  | 6  |
|     |                                                                                                                                        |                                                       |                                                        |                                                                        |  |    |
| Q.3 | Solve the following.                                                                                                                   |                                                       |                                                        |                                                                        |  | 12 |
| A)  | Discuss the different types of turbines used in hydroelectric power plants. Explain their working principles and suitable applications |                                                       |                                                        |                                                                        |  | 6  |
| B)  | Draw layout & explain the working principle of a gas turbine power plant.                                                              |                                                       |                                                        |                                                                        |  | 6  |

|                    |                                                                                                                                                                                                                                                                                                                                                                                           |  |           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|
|                    |                                                                                                                                                                                                                                                                                                                                                                                           |  |           |
| <b>Q. 4</b>        | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                    |  | <b>12</b> |
| <b>A)</b>          | Explain with a schematic diagram of a nuclear power plant.                                                                                                                                                                                                                                                                                                                                |  | <b>6</b>  |
| <b>B)</b>          | Discuss with neat diagram the design and operation of a pressurized water reactor (PWR)                                                                                                                                                                                                                                                                                                   |  | <b>6</b>  |
| <b>C)</b>          | Explain with a neat diagram the working principle of a diesel power plant.                                                                                                                                                                                                                                                                                                                |  | <b>6</b>  |
|                    |                                                                                                                                                                                                                                                                                                                                                                                           |  |           |
| <b>Q. 5</b>        | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                    |  | <b>12</b> |
| <b>A)</b>          | Explain with a neat diagram the working principle of Magneto-hydrodynamic (MHD) power generation. Discuss the key components of an MHD generator                                                                                                                                                                                                                                          |  | <b>6</b>  |
| <b>B)</b>          | Explain the working principle of a wind turbine power plant. Include a labelled diagram of a horizontal-axis wind turbine (HAWT) and describe the following: <ol style="list-style-type: none"> <li>1. The role of each major component (blades, rotor, generator, yaw mechanism, etc.).</li> <li>2. How energy is converted from kinetic energy of wind to electrical energy.</li> </ol> |  | <b>6</b>  |
| <b>C)</b>          | Design a layout for a solar photovoltaic (PV) power plant and explain in detail.                                                                                                                                                                                                                                                                                                          |  | <b>6</b>  |
|                    |                                                                                                                                                                                                                                                                                                                                                                                           |  |           |
| <b>Q. 6</b>        | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                    |  | <b>12</b> |
| <b>A)</b>          | How do combined power plants contribute to the stability of an electrical grid during sudden fluctuations in power demand? Explain the role of spinning reserves and reserve margins in maintaining grid stability.                                                                                                                                                                       |  | <b>6</b>  |
| <b>B)</b>          | Discuss the advantages and disadvantages of combining renewable energy sources (such as solar or wind) with conventional power plants (such as thermal or nuclear) in a hybrid power generation system.                                                                                                                                                                                   |  | <b>6</b>  |
| <b>C)</b>          | Describe the concept of "Base Load" and "Peak Load" in the context of a combined operation of power plants. How can thermal, hydro, and renewable plants be optimized for these load types?                                                                                                                                                                                               |  | <b>6</b>  |
| <b>*** End ***</b> |                                                                                                                                                                                                                                                                                                                                                                                           |  |           |